

Abdur Rahman



EDUCATION

King Fahd University of Petroleum and Minerals (3.625/4.00)

Dhahran, Saudi Arabia

MS in Applied Statistics

2023-2025

- Thesis: Geospatial and Machine Learning Mortality Modeling of MENA Countries

The Islamia University of Bahawalpur (3.93/4.00)

Bahawalpur, Pakistan

Bachelor of Statistics

2018-2022

- Ranked first in cohort (88.44%)

PUBLICATIONS

Peer-reviewed / Published

[On designing EWMA charts to model demographic trends with generalized extreme value distribution](#)

Abdur Rahman, M. Hafidz Omar, Nasir Abbas, Muhammad Riaz | Mathematical Population Studies (Routledge)-2026

[Modeling and forecasting air pollution for public health protection based on ML and time series models in GCC countries](#)

Muhammad Daniyal, Hanan Alyami, Ali Jaber Alqahtani, Muhammad Talha, Abdur Rahman, Abbas Al Mutair | BMC Public Health, 26, 280 (Springer Nature)- 2026

[Water level forecasting in coastal cities using a hybrid deep learning approach](#)

Abdur Rahman, M. Hafidz Omar, Tahir Mahmood, Nasir Abbas, Muhammad Riaz, Naeem Ramzan | Science of the Total Environment, 1003, 180709 (Elsevier)- 2025

Conference Proceedings

[On Designing EWMA Charts to Model Demographic Trends with Generalized Extreme Value Distribution](#)

Abdur Rahman, M. Hafidz Omar, Nasir Abbas, Muhammad Riaz | 39th International Workshop on Statistical Modelling (IWSM 2025), Limerick, Ireland

Submitted / Under Review / In Preparation

[Optimizing Demographic Surveillance using GEV Control Charts: An Advanced Approach for Monitoring Fertility Trends](#)

Abdur Rahman, M. Hafidz Omar, Nasir Abbas, Muhammad Riaz | Submitted

[A Generalized Extreme Value CUSUM Framework for Detecting Shifts in Infant Mortality Trends: Advancing Public Health Surveillance](#)

Abdur Rahman, Nasir Abbas, M. Hafidz Omar, Muhammad Riaz | Under Review

[Hybrid Residual Learners Framework for Mortality Forecasting: Integrating Lee-Carter, Renshaw-Haberman and Machine Learning Models in the MENA Region](#)

Abdur Rahman, M. Hafidz Omar | Submitted

[A Hybrid Deep Learning Architecture for Multi-Dimensional SDG Modelling](#)

In preparation

[Learning Spatial Structure in Mortality Data: ESF vs Modern Machine Learning](#)

In preparation

HONOURS AND AWARDS

KFUPM Merit-Based MS Scholarship (2023)

- King Fahd University of Petroleum and Minerals competitive, merit-based scholarship

Gold Medal (2022)

- The Islamia University of Bahawalpur, Pakistan awarded for ranking first in cohort, BS Statistics

Ehsaas Undergraduate Scholarship (2020-2022)

- Higher Education Commission (HEC), Government of Pakistan merit-based national scholarship covering semester tuition fees and education-related expenses

PEEF Merit Scholarship (2016)

- Punjab Educational Endowment Fund (PEEF), Government of Punjab awarded for academic achievement in the Intermediate examination

RESEARCH EXPERIENCE

Research Mentor

2024 – present

King Fahd University of Petroleum & Minerals

Guiding four junior researchers Fatima Zahra Hawwa, Jafar Radhi Abu Qurayn, Tahir Ali, and Muhammad Zakariya on independent research projects in statistical modeling and machine learning, combining formal and informal advisory mentoring. Two resulting manuscripts are currently in preparation for submission, and a third collaboration is at the results stage.

Graduate Researcher (MS Thesis)

2023 – 2025

King Fahd University of Petroleum & Minerals

Thesis: Geospatial and Machine Learning Mortality Modeling of MENA Countries integrated statistical process control and deep learning for demographic and environmental forecasting; work led to three peer-reviewed publications.

TEACHING EXPERIENCE

R Language Lab Instructor

2024 – 2025

King Fahd University of Petroleum & Minerals

Python Language Lab Instructor

2024

King Fahd University of Petroleum & Minerals

PROFESSIONAL AND FIELD EXPERIENCE

Teacher Assistance

2023 – 2024

King Fahd University of Petroleum & Minerals

Survey Enumerator

2022

Bureau of Statistics Punjab, Bahawalpur, Pakistan

LEADERSHIP AND EXTRACURRICULAR

Class Representative

2021 – 2022

The Islamia University of Bahawalpur — Faculty of Computing, Department of Statistics

Recognized with a Certificate of Appreciation for outstanding performance as Class Representative, coordinating between faculty and peers throughout the academic year.

CONFERENCES AND SEMINARS

39th International Workshop on Statistical Modelling 2025

Limerick, Ireland

GSR Conference 2025

Math Connect 2024

VOLUNTEERING

Social Service — Dhahran, Saudi Arabia

Arabian Warrior OCR — Dhahran, Saudi Arabia